Lecture 1(a) : Introduction to power system

Dr. Md Umar Hashmi

KU Leuven & EnergyVille, Belgium

Who are we?

KU LEUVEN

Internationally acclaimed research university

KU Leuven founded in 1425 completed 600 years of continuous operation last year.

KU Leuven is consistently in the **top 50** in global rankings –
every faculty ranks in the top-100 in its academic discipline.

BELGIUM

EUROPE

WORLD

Times Higher Education Ranking (2026)

A vibrant university **community**

University Staff

13,000+

Students

64,000+

PhD students

7,000+

Alumni

329,000+

Incoming exchange students

2,000+

A diversified international student body

16,000+ International Students

140+ Nationalities

35+ Agreements with Top-50 universities

Our research in numbers

YEARLY RESEARCH EXPENSES

750

MILLION EUROS

RESEARCHERS

10,375

Academic articles

10K+

PER YEAR

ERC GRANTS

168

SINCE 2007

PHD DEGREES AWARDED

1023

2023-2024

KU Leuven fosters **entrepreneurship** and transforming **innovative ideas** into successful business models and start-ups.

PATENTS GRANTED

131

(IN 2024)

SPIN-OFFS CREATED

186

EnergyVille -- a brief overview

- What is EnergyVille? A collaboration formalized in a Memorandum of Understanding “EnergyVille” since 2018

KU LEUVEN vito imec UHASSELT

- More than 500 people involved
- Over 400 researchers (about 50% international)
- 8 research topics

Battery storage	Electrical networks	Energy for buildings and districts	Energy strategies and markets	Power electronics	Power-to-Molecules	Solar energy	Thermal systems

Campus EnergyVille

Introduction – Md Umar Hashmi

2010 - 2012

M.Tech. – Indian Institute of Technology Bombay

- Energy Systems Engineering

2012 - 2015

Control engineer at Eaton Corporation

- Contributed towards modelling and simulation of electrical, hydraulic and aerospace projects
- Holds a US patent as the primary inventor

2015 - 2019

PhD: École Normale supérieure, PSL University, France

- Optimization and control of energy storage devices
- Spent 3 and 6 months at Los Alamos National Laboratory and Madeira Institute of Technology respectively

2020

Senior Engineer Planning, Operation and Innovation, Eirgrid

- Assessing the impact of Celtic interconnector of Ireland

2020 - 2023

Postdoctoral researcher – KU Leuven & EnergyVille

- Decision support tools for system operators in distribution networks
- Dynamic operating envelopes for Moonshot

2023 - ....

Research expert– KU Leuven & EnergyVille

- Decision support tools for system operators in distribution networks
- Contributing to distribution network roadmap and team

C2-FlexIQ

Course details

What to expect?

Course composition

- 13 Lectures
  - 19 hours of lectures: D1: 4.5 hr, D2: 2.5 hr, D3: 4.5 hr, D4: 4.5 hr, D5: 3 hr
- 7 academic instructors
  - 7 lectures by myself
  - 2 by Prof. Geert Deconinck
  - 1 each by Prof Dirk Van Hertem, Prof Hussain Kazmi, Dr. Mohamed Bahloul, Dr. Jelle Meus & Elise Chambard, Dr. Deep Deka
- 9 industrial talks
  - ELIA, Fluvius, YUSO, Octave, Centrica, CREG, VITO, N-Side
- 2 Teaching assistants
  - Alexandre Venâncio Gouveia
  - Elias Mandefro Getie
- 3 Hands on exercise sessions

Course goals & structure

- Day 1 → General introduction to power systems, & flexibility: definition, classification and its growing needs
- Day 2 → Introduction to power system optimization, power flow and fairness with hands on implementation of the energy arbitrage problem
- Day 3 → Smart metering as flexibility enablers and flexibility quantification: theory and implementation
- Day 4 → Network awareness and flexibility markets
- Day 5 → Flexibility management, DER identification and alternative for flexibility

Course acknowledgements

Hossein

Alexandre

Elias

Ali

Zijie

Leander

Milan

Olivia

Dirk

Geert

Pedro

Aiko Fias

Manna Carlo

Matias

Deep Deka

Ana Busic

Anne

Contact details

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  - Alexandre Venâncio Gouveia -- alexandremiguel.venanciogouveia@kuleuven.be
  - Elias Mandefro Getie -- eliasmandefro.getie@kuleuven.be

Day assignments

- ***Day Assignment 1*** – flexibility value chain stakeholders in your allocated country with examples of national stakeholders: DSO, TSO, aggregator, market operator, flexibility provider, consumers, prosumers.
- ***Day Assignment 2*** – Discuss smart metering penetration, meter types, data measured (P, Q, V, I, frequency etc.), data exchange frequency, communication mediums used in your allocated country
- ***Day assessment 3*****:** How much flexibility potential is needed at your allocated country level with large scale DER adoption?
- ***Day assignment 4*****:** Compare grid investment needs vs flexibility planning cost in your allocated country?

The day assignments are expected to be 1–2-page writeups on the topic.

Day assignments are based on your allocated country

Last name	First name	Country allocated
ALEXANDER	LUCY LYUDMILA	Austria
CARDOSO	Beatriz	Bulgaria
DACREMA	Sara	Croatia
DI BELLA	Alessandro	Cyprus
DUTÝ	Šimon	Czechia
ENACHE	Andreea-Irina	Denmark
ERDOĞDU	Esra	Estonia
ESTÉVEZ MARÍN	DIEGO	Finland
FERRER SCHAAP	Noah Vicent	France
GIRELLI	Oscar	Germany
HÂRSEANU	Bianca	Greece
JAKUBOWSKI	Piotr	Hungary
KAYAOĞLU	Emir Faruk	Ireland
KRASNODĘBSKI	Dominik	Italy
KULIGOWSKI	Adam	Latvia
MORENO GIL	Andrea	Poland
PIZOŃ	Aleksander	Luxembourg
ROMAN RUIZ	Mercedes	Malta
SALVATIERRA DE LA CRUZ	Gianmarco	Netherlands
SÜT	Özer	Norway
TOPOLNICKI	Łukasz	Portugal
TRAVAGLIA	Alice	Romania
TREZZINI	Federico	United Kingdom
URRUTIA VÁSQUEZ	Joel Adrián	Slovenia
ZDZIARSKI	Szymon	Spain
ÖRÜN	Erencan	Sweden

Final assignment → topics

Final assignment: write approx. 4000 words on any of the following allocated topics:

1. Stakeholder engagement for harnessing flexibility value chain: towards fairness
   - Stakeholders to consider → TSO, DSO, market operator, passive consumers, active consumers, flexible assets, aggregators
2. Pathways to harnessing distributed flexibility: How should users be encouraged to offer flexibility?
   - Use real world examples for making a hypothesis
3. A future towards consumer centric power distribution network and the role of other stakeholders
4. Flexibility adequacy → Summarize key challenges with real world or your country's specific examples of why we would need flexibility
5. Distribution customer empowerment – a global perspective towards democratizing active consumer behavior

Alternate topics can be discussed and agreed upon; All such discussions should be done prior to picking a topic.

Final assignment → submission guidelines

Structuring your work with following considerations

- The write-up should discuss (i) national policies, (ii) European regulation, (iii) legal barriers, (iv) privacy barriers, (v) locational fairness, ...
- Structure the write up in terms of dimensions of (i) technology enablers, (ii) policy enablers, (iii) regulatory enablers
- Use real world examples for making hypothesis
- Proper referencing is mandatory. Follow best practices for scientific writing

What is expected:

- The use of AI in assignments should be limited to grammatical corrections.
- The document should be reference properly to the documents
- Students should prepare the document in Overleaf and share the link
- Overleaf folder should be named as: PSF_FullName (Full name without space)
  - Tutorial on overleaf → https://www.youtube.com/watch?v=xcTN4F3l9Ds

Course grading

- Day 1 to 4 assignments for 5 points each
  - Deadline to submit → **29th March 2026**

- Final essay for 80 points
  - Deadline to submit → **12th April 2026**

Course acknowledgements

Kennis kent geen einde

IMPROCAP

https://www.improcap.eu/

DiGiRES

https://www.digi-res.eu/

C2-FlexIQ

https://research.kuleuven.be/portal/nl/project/3E240512

KU LEUVEN

Energy Ville ENERGY IN TRANSITION

LEUVEN INSTITUTE ENERGY SOCIETY

VLAIO

flux50 CATALISTI

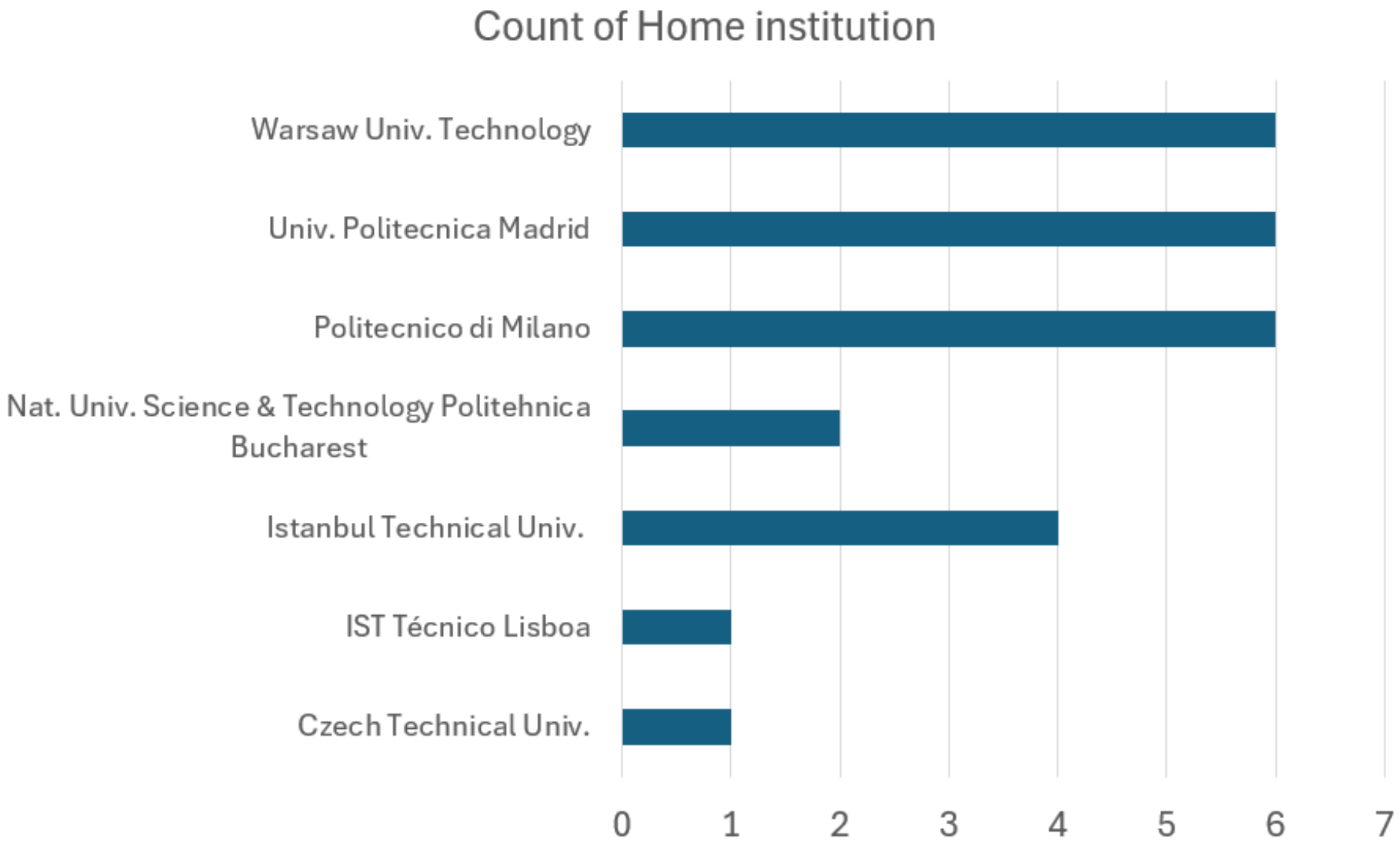
fwo

MITei MIT Energy Initiative

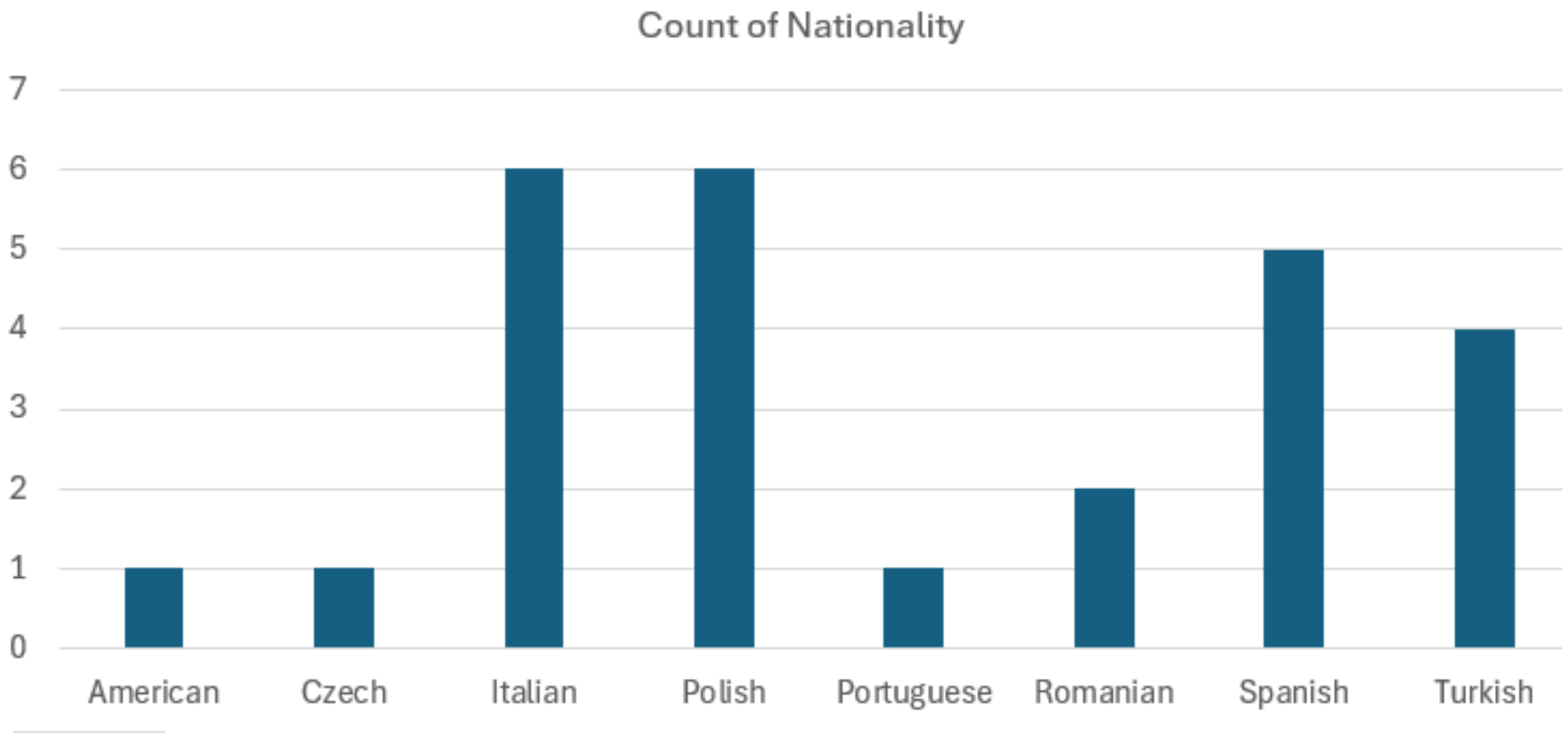
Class composition

What to expect?

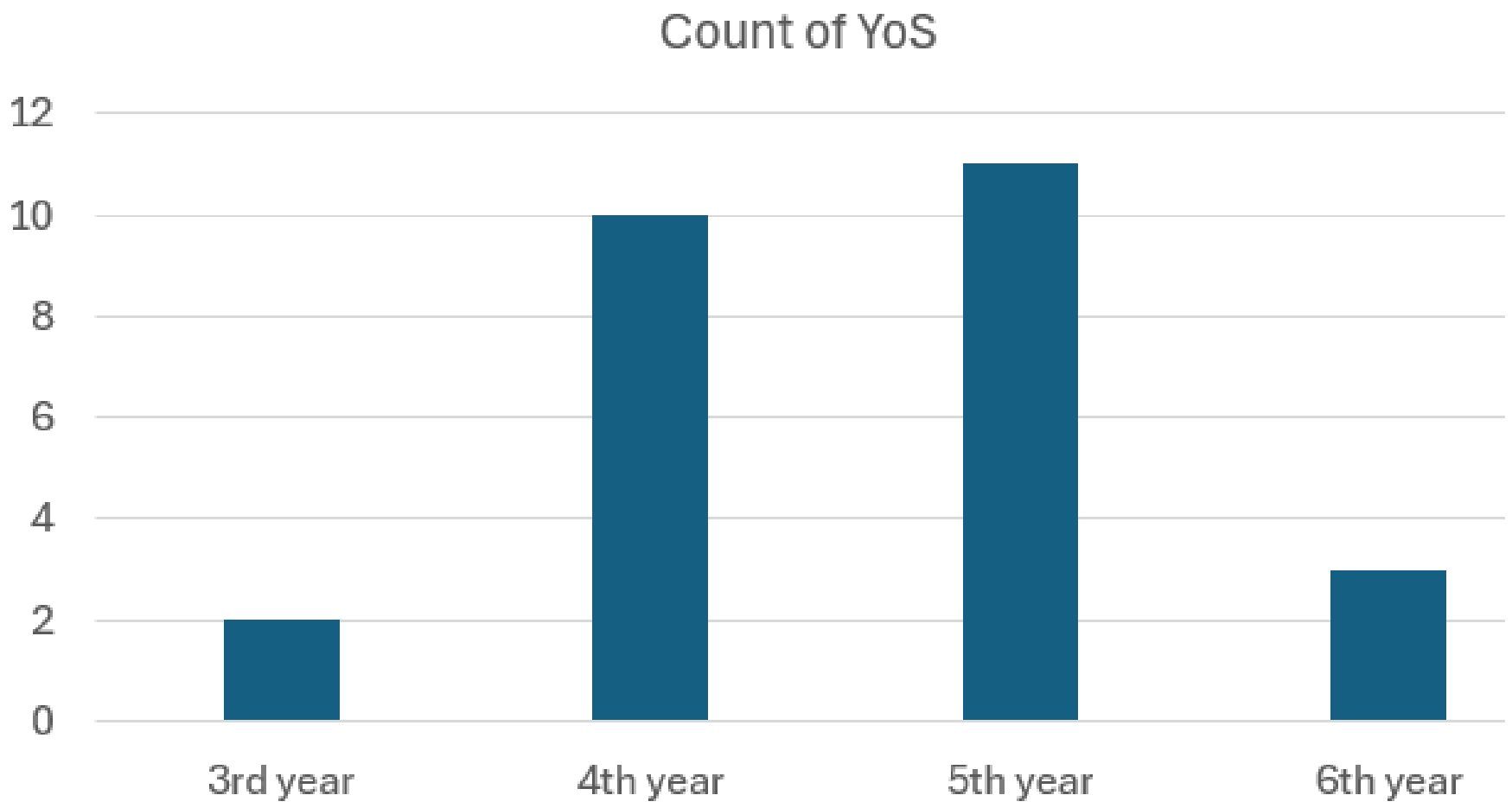
Class composition – Home institute



Class composition – Nationality



Class composition – Year of study



Are you aware of flexibility in Power System?

Yes

0%

No

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How many of you know what a TSO and DSO are?

Yes

0%

No

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Do you know what reactive power is

Yes

0%

No

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Have you implemented an optimization problem

Yes

0%

No

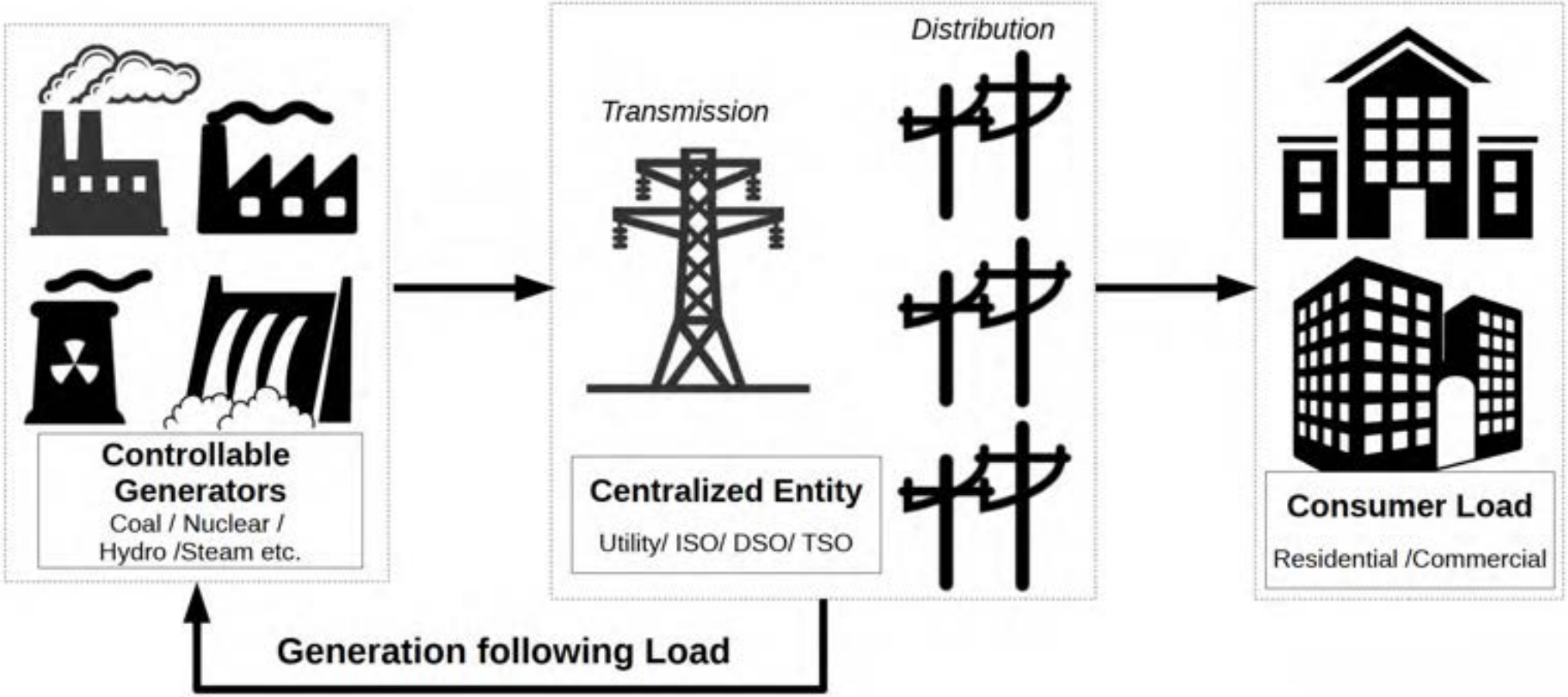
0%

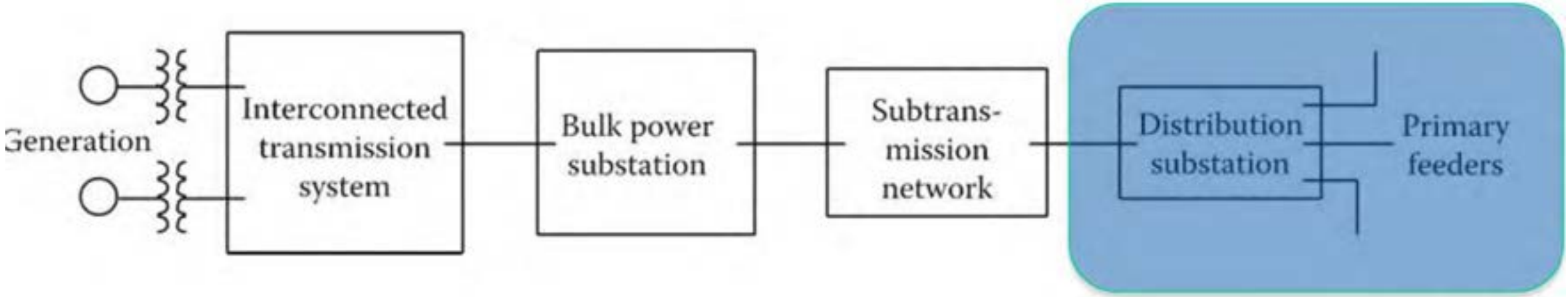
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Traditional operation of power networks

Conventional power system

Decoupled generation & consumption





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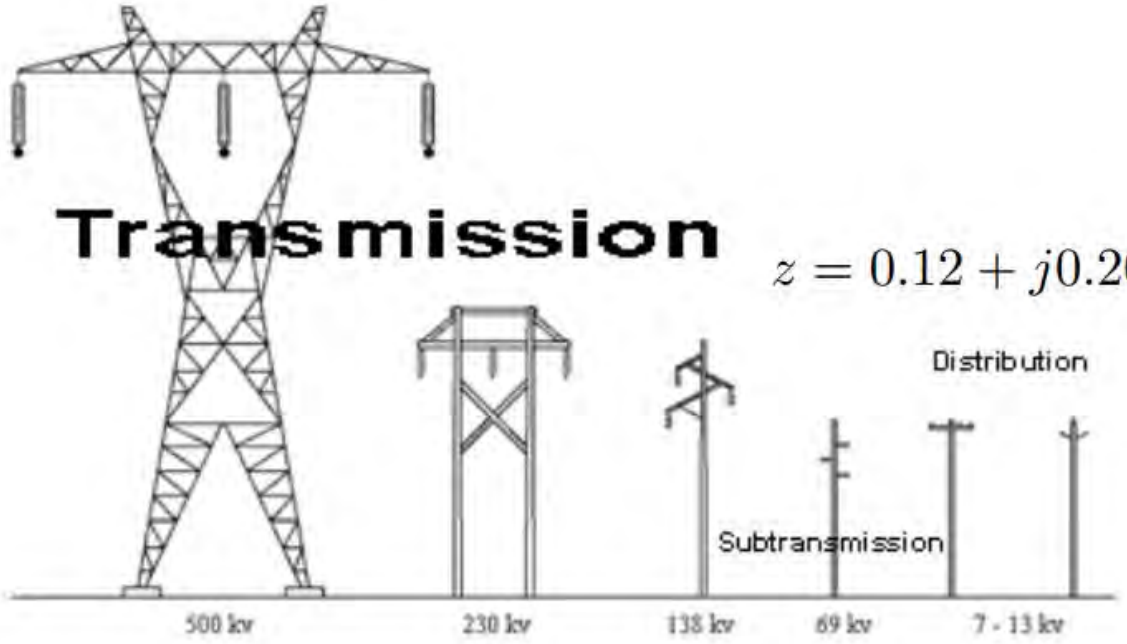


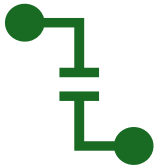
$z = 0.12 + j0.20\ \Omega/\text{mile}$

$z = 0.05 + j0.5\ \Omega/\text{mile}$

Courtesy: V. Kekatos, Purdue

Power distribution network:

Helps in connecting electricity consumers
using (often) radial (spanning tree) networks



Distribution networks (DNs) consist of

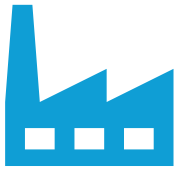
Distribution transformer

Cables connecting power source with consumers

- Topologically comprising of nodes and branches of the graph

Electricity consumers, prosumers, distributed energy resources (DERs)

Protection devices etc.



Entities involved in distribution network operation

Distribution system operator (DSO): ensures reliable operation of the DN

Consumers/Prosumers : consumes/generates electrical energy includes DERS, flexibility etc.

Aggregator: resource aggregator for upstream grid services such as ancillary services

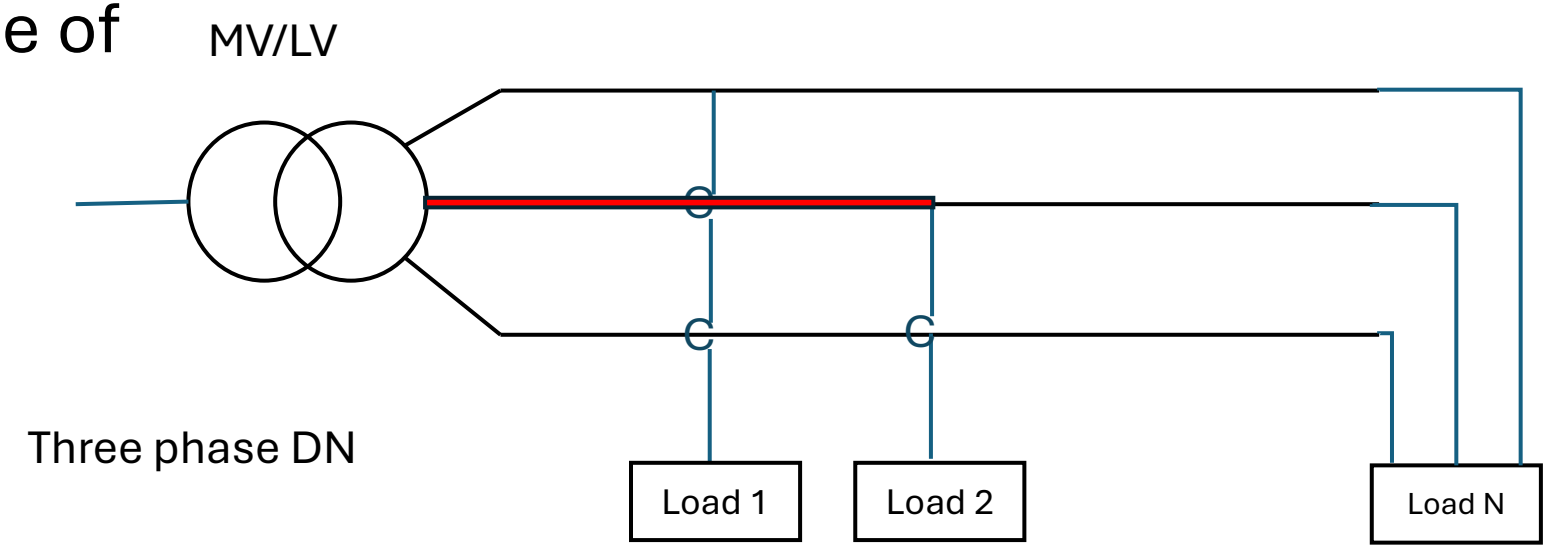
Market operator, Retailer,

What are **distribution network incidents** (DNIs)

DNIs primarily compose of

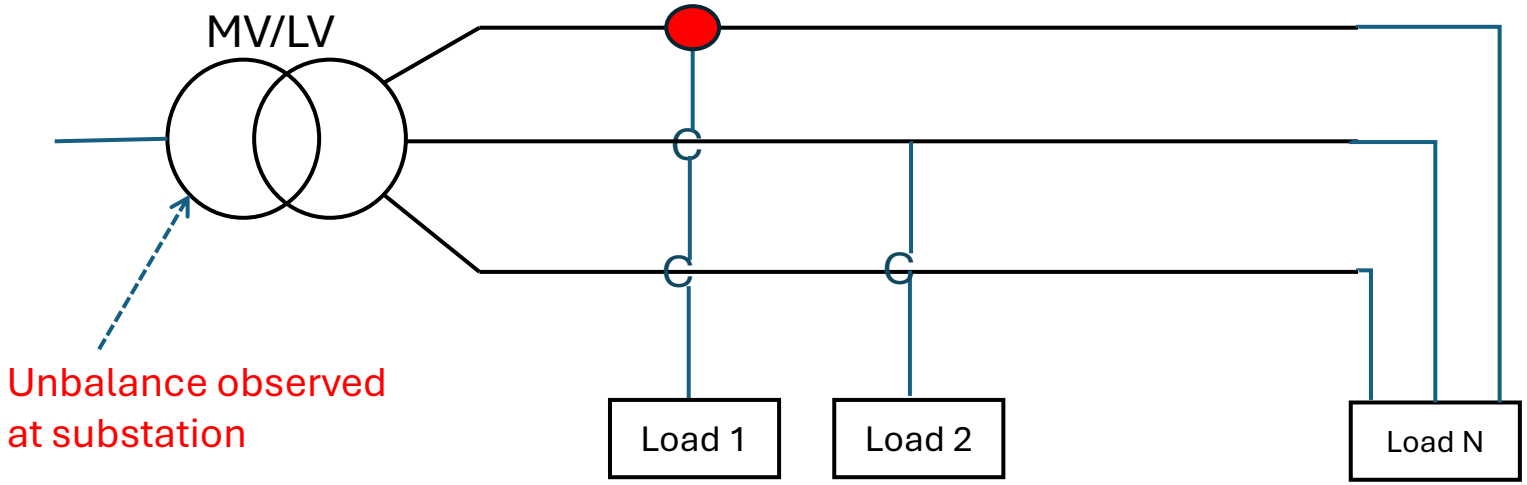
- Line overloads

Branch attributes ➔



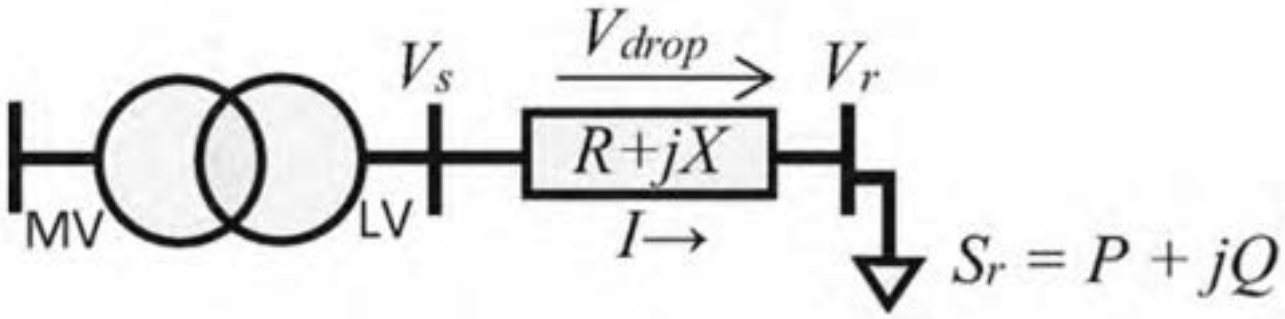
- Power quality limit violations: voltage, unbalance, harmonics etc.

Node attributes ➔



Electric power flow

S	=	Apparent power
V	=	voltage magnitude
I	=	Line current
P	=	Active power
Q	=	Reactive power
R	=	line resistance
X	=	line reactance
Z	=	line impedance



Simple lumped-element circuit model for LV regulation (Marinelli 2025a)

$$S_r = V_r I^* = P + jQ \Rightarrow I = \frac{(P+jQ)^*}{V_r^*} = \frac{P - jQ}{V_r}$$

$$V_s = V_r + V_{drop} = V_r + ZI = V_r + (R + jX)\frac{(P - jQ)}{V_r}$$

$$V_{drop} = V_s - V_r = (R + jX)\frac{(P - jQ)}{V_r} = \frac{(PR + QX)}{V_r} + j\frac{(PX - QR)}{V_r}$$

$$V_{drop} \cong \frac{(PR + QX)}{V_r}$$

Power quality metric classification

Power quality problems may be classified on the basis of events such as **transient and steady** state, the **quantity** such as current, voltage, and frequency, or the **load and supply systems, entity** and **voltage level**. Other classification criteria: (a) **ownership,** (b) **reliability**, (c) **new interconnection rule** etc.

On the basis of **TIMESCALE**

- Steady state
  - Long-duration voltage variations,
  - Waveform distortions,
  - Unbalanced voltages,
  - DC offset,
  - Flicker,
  - Poor power factor,
  - Unbalanced load currents,
  - Load harmonic currents,
  - Excessive neutral current
- Transient state
  - Voltage sag (dip),
  - Swell,
  - Short-duration voltage variations,
  - Power frequency variations, and
  - Voltage fluctuations

On the basis of **QUANTITY**

- Voltage
  - Voltage distortions,
  - Flicker,
  - Sag,
  - Swell,
  - Unbalance,
  - Under-voltage/ overvoltage
- Current
  - Reactive power component of the current,
  - Harmonic currents,
  - Unbalanced currents,
  - Excessive neutral current, etc.
- Frequency

Based on **VOLTAGE LEVEL**

(same classification for transmission / distribution)

- High voltage
  - Frequency
  - Voltage
- Medium and Low voltage
  - Voltage flicker
  - Harmonics
  - Reactive Power
  - Voltage rise/dip
  - Power transition

Based on **ENTITY**

- Generators
  - Large, small, medium size
  - Coal, nuclear, renewable
- Industrial/commercial/ domestic prosumers

Based on **LOAD OR SUPPLY** system

- Problems due to load
  - Current harmonics,
  - Reactive power component of the current,
  - Unbalanced currents,
  - Neutral current,
  - DC offset
- Problems due to supply
  - Voltage and frequency related issues such as voltage distortion,
  - Unbalance,
  - Sag,
  - Swell & flicker

EN 50160

Power quality in rules for European Union

To consider for steady state analysis

Supply voltage phenomenon	Acceptable limits	Measurement Interval	Monitoring Period	Acceptance Percentage
Grid frequency	49.5Hz to 50.5Hz 47Hz to 52Hz	10 s	1 Week	95% 100%
Slow voltage changes	230V ± 10%	10 min	1 Week	95%
Voltage Sags or Dips (≤1min)	10 to 1000 times per year (under 85% of nominal)	10 ms	1 Year	100%
Short Interruptions (≤ 3min)	10 to 100 times per year (under 1% of nominal)	10 ms	1 Year	100%
Accidental, long interruptions (> 3min)	10 to 50 times per year (under 1% of nominal)	10 ms	1 Year	100%
Temporary over-voltages (line-to-ground)	Mostly < 1.5 kV	10 ms	N/A	100%
Transient over-voltages (line-to-ground)	Mostly < 6kV	N/A	N/A	100%
Voltage unbalance	Mostly 2% but occasionally 3%	10 min	1 Week	95%
Harmonic Voltages	8% Total Harmonic Distortion (THD)	10 min	1 Week	95%

Source: http://www.ecoca.ro/ntponeusvro/en50160.pdf EN 50160:2000 Power Quality Fact Sheet prepared by **www.reo.co.uk**

Voltage magnitude limits

Synchronous Area	Voltage Range	Time period for operation
Continental Europe	0.85 pu – 0.90 pu	60 minutes
	0.90 pu – 1.118 pu	Unlimited
	1.118 pu – 1.15 pu	To be decided by each TSO while respecting the provisions of Article 4(3), but not less than 20 minutes
Nordic	0.90 pu – 1.05 pu	Unlimited
	1.05 pu – 1.10 pu	60 minutes
Great Britain	0.90 pu–1.10 pu	Unlimited
Ireland	0.90 pu – 1.118 pu	Unlimited
Baltic	0.85 pu – 0.90 pu	30 minutes
	0.90 pu – 1.12 pu	Unlimited
	1.12 pu – 1.15 pu	20 minutes

Table 6.1: This table shows the minimum time periods a Power Generating Module shall be capable of operating for Voltages deviating from the nominal value at the Connection Point without disconnecting from the Network. (The Voltage base for pu values is from 110 kV to 300 kV (excluding).)

Synchronous Area	Voltage Range	Time period for operation
Continental Europe	0.85 pu – 0.90 pu	60 minutes
	0.90 pu – 1.05 pu	Unlimited
	1.05 pu – 1.0875 pu	To be decided by each TSO while respecting the provisions of Article 4(3), but not less than 60 minutes
	1.0875 pu – 1.10 pu	60 minutes
Nordic	0.90 pu – 1.05 pu	Unlimited
	1.05 pu – 1.10 pu	60 minutes
Great Britain	0.90 pu – 1.05 pu	Unlimited
	1.05 pu – 1.10 pu	15 minutes
Ireland	0.90 pu – 1.05 pu	Unlimited
Baltic	0.88 pu – 0.90 pu	20 minutes
	0.90 pu – 1.10 pu	Unlimited
	1.10 pu – 1.15 pu	20 minutes

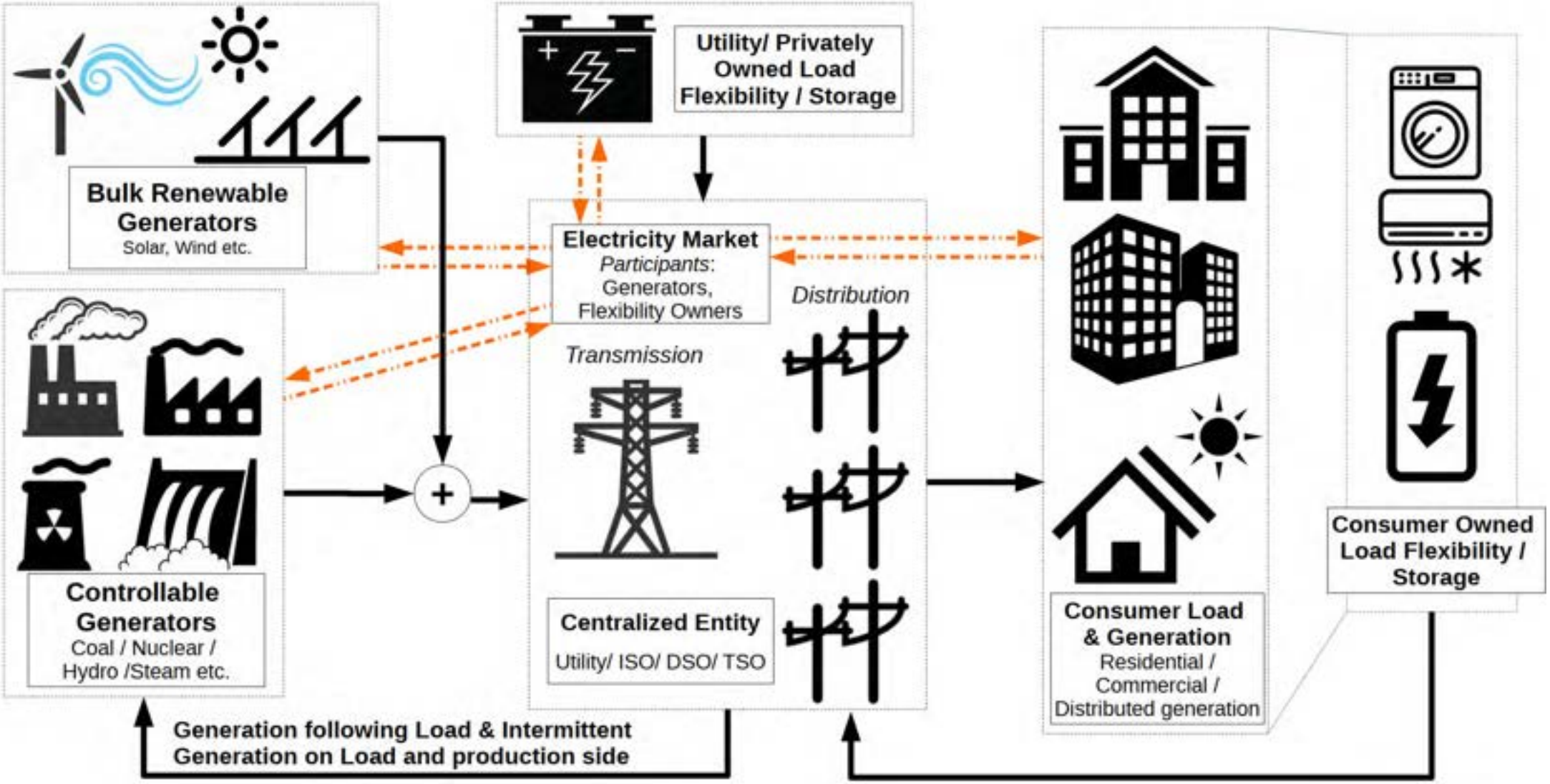
Table 6.2: This table shows the minimum time periods a Power Generating Module shall be capable of operating for Voltages deviating from the nominal value at the Connection Point without disconnecting from the Network. (The Voltage base for pu values is from 300 kV to 400 kV.)

Source: ENTSO-E Network Code for Requirements for Grid Connection Applicable to all Generators
https://eepublicdownloads.entsoe.eu/clean-documents/pre2015/resources/RfG/130308_Final_Version_NC_RfG.pdf

Evolution of power networks, consumption patterns

New power system

Widespread intermittency in consumption & generation



New challenges in power system

1 INTERMITTENCY

- Greater intermittency in load and generation
- Increased penetration of DERs

Greater variability in load and generation increases system uncertainty

2 SIMULTANEITY

- High simultaneity in intermittency
- Due to increased consumer rationality
- Weather/behavioural similarity in load/generation

High coincident demand worsens intermittency and stresses the grid.

3 MARKET DESIGN GAPS

- Lack of incentives for harnessing distributed flexibility
- Mismatch between grid need and market design

Weak incentives and misaligned markets limit flexible resource use.

4 CONNECTION DELAYS

- Bottleneck in power networks – delaying new connections
- Ageing infrastructure and high DER penetration create capacity scarcity, forcing DSOs to delay new connections.

Grid bottlenecks and ageing assets slow new customer connections.

5 LACK OF COORDINATION

- Lack of coordination among key stakeholders – TSO, DSO, asset owners, aggregator, market design
- Under utilization of resources

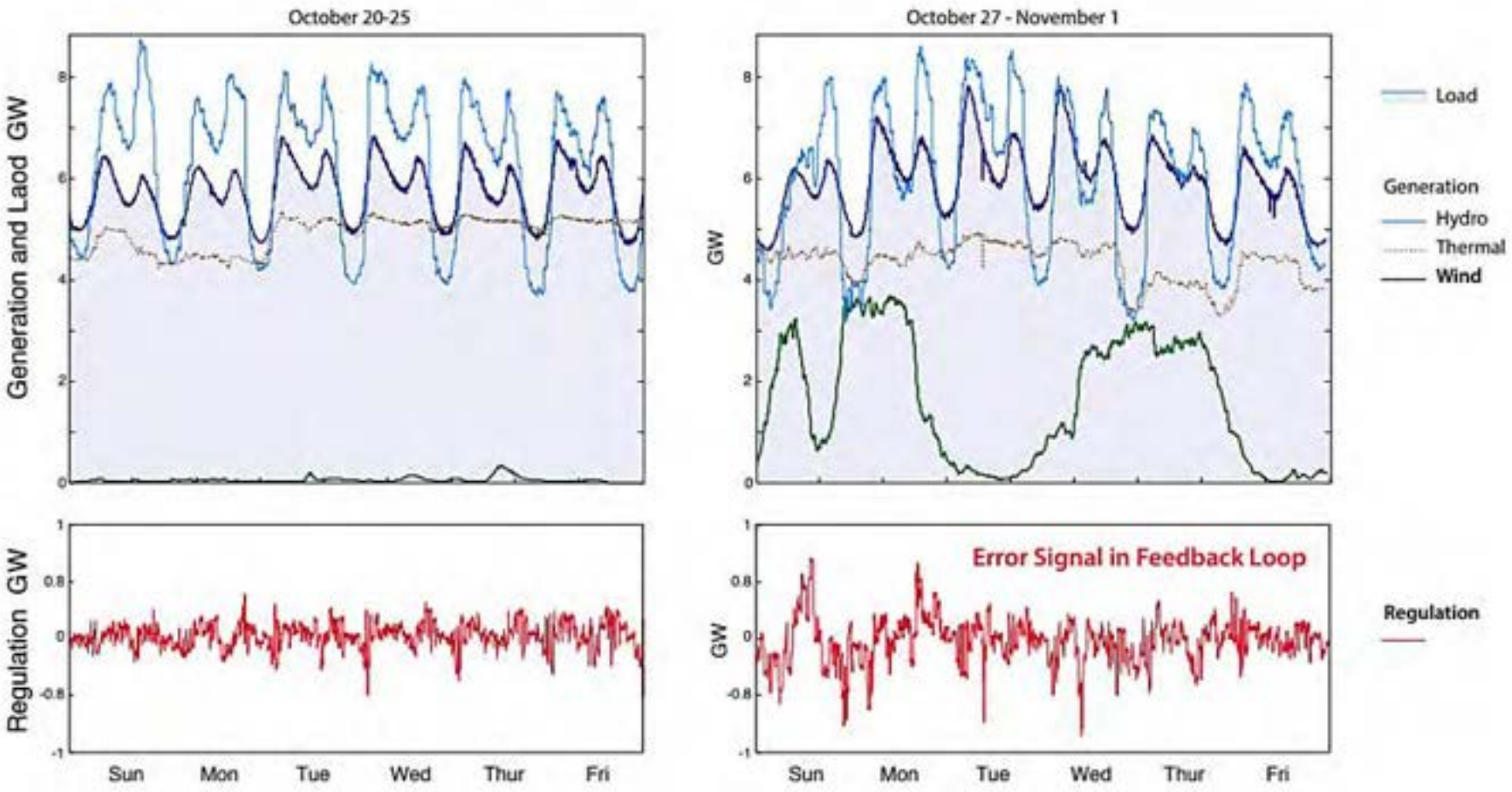
Poor stakeholder alignment leads to inefficient grid operations.

6 OBSERVABILITY

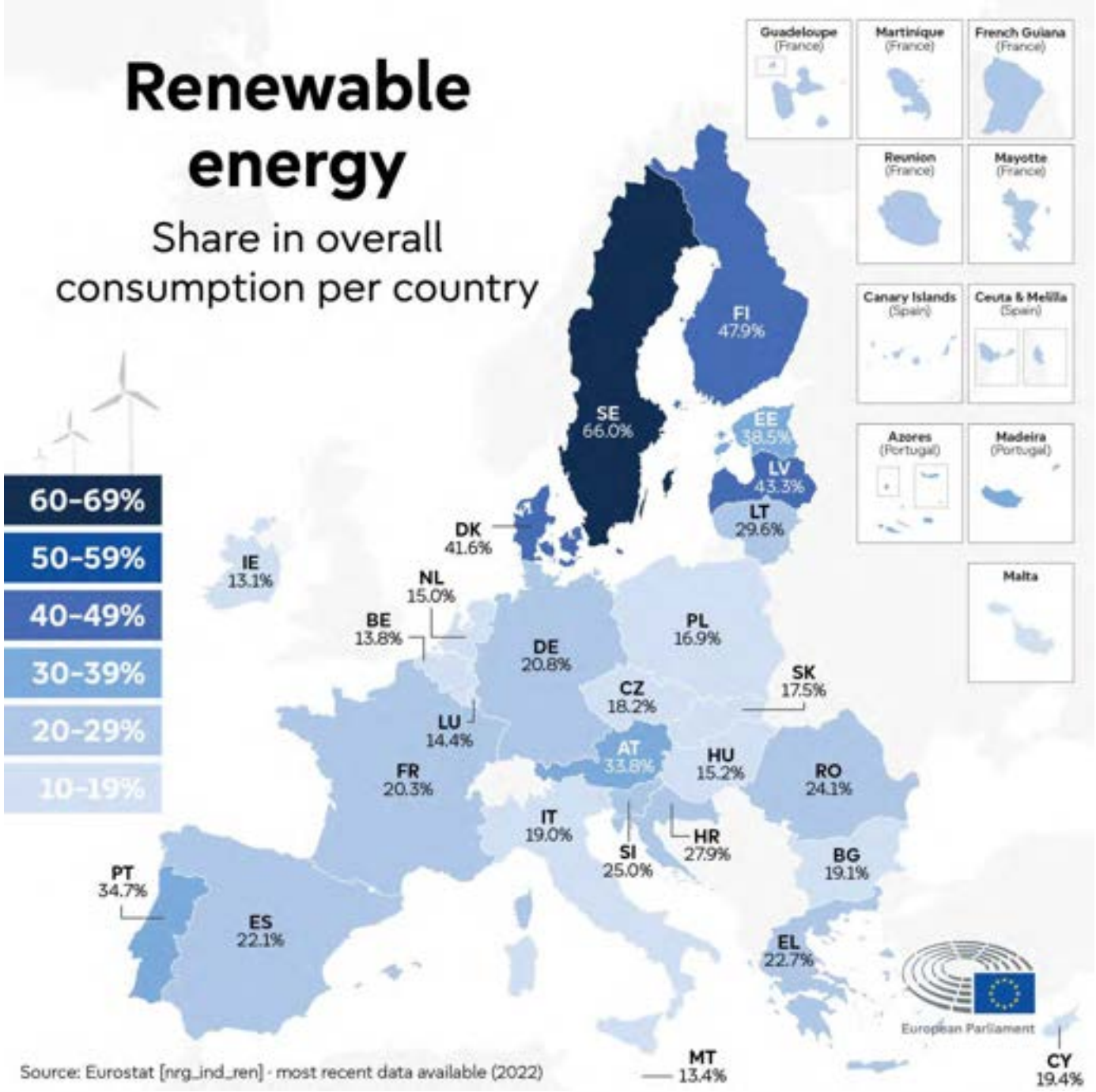
- Asset visibility and observability limiting controllability
- Privacy and data exchange concerns
- Need for technology, policy, regulatory enablers

Limited visibility and data access restrict effective grid control.

1. Greater intermittency in load and generation

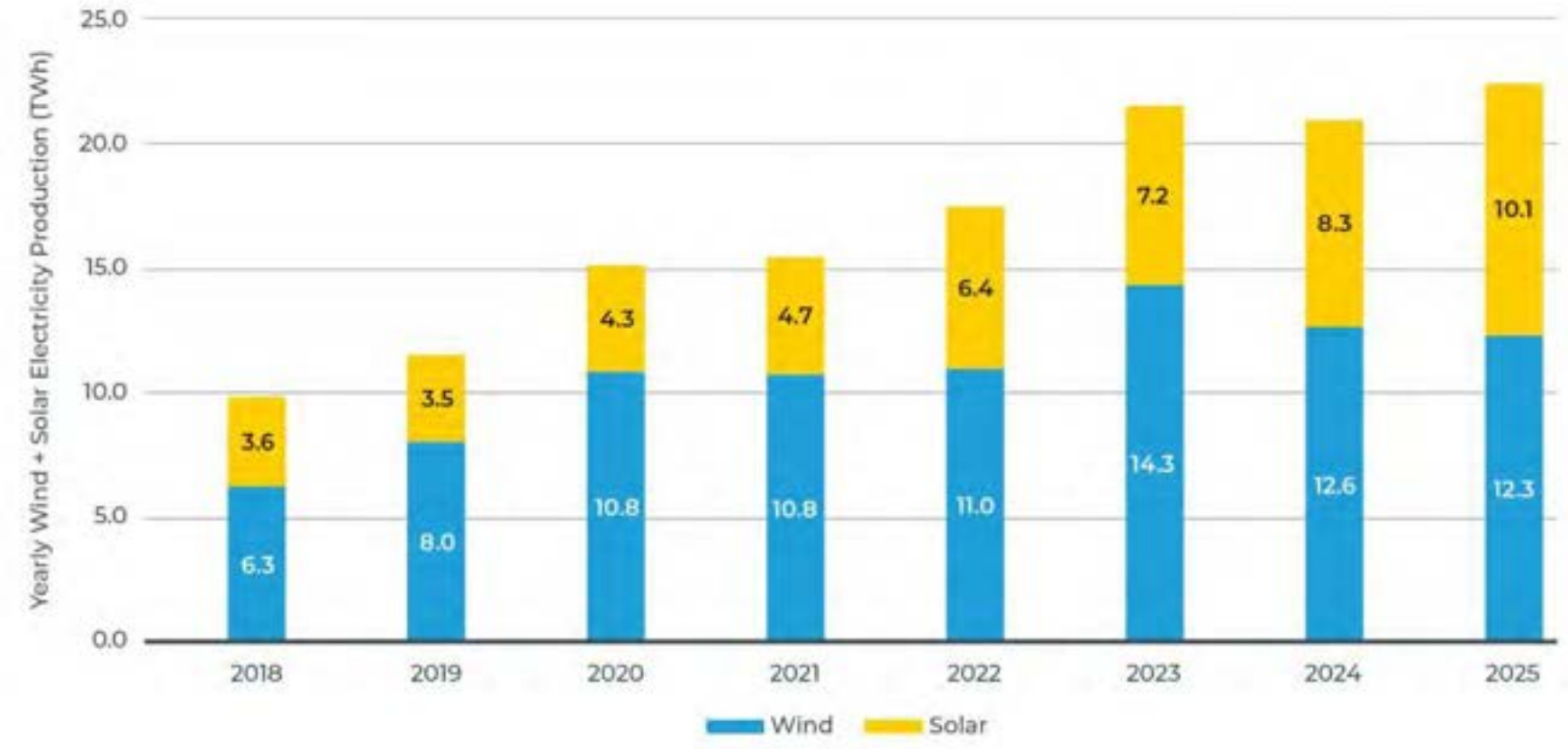


RES in Europe



Source:
https://www.europarl.europa.eu/topics/en/article/20221128STO58001/how-the-eu-is-boosting-renewable-energy

Electricity Generated from Wind + Solar Power (TWh)



Belgian electricity consumption remained stable at 80.1 TWh.

Source: https://www.elia.be/en/newsroom/2025/12/20251226_press-release-electricity-mix

New loads: EV charging & Heat pump load

- The sales have increased by **+50%** in 2022, compared to 2021 for the EVs
- The sales for heat pumps have also increased by **+60%** in 2022 compared to 2021

- High peaks
- Highly simultaneous

2. High simultaneity in intermittency

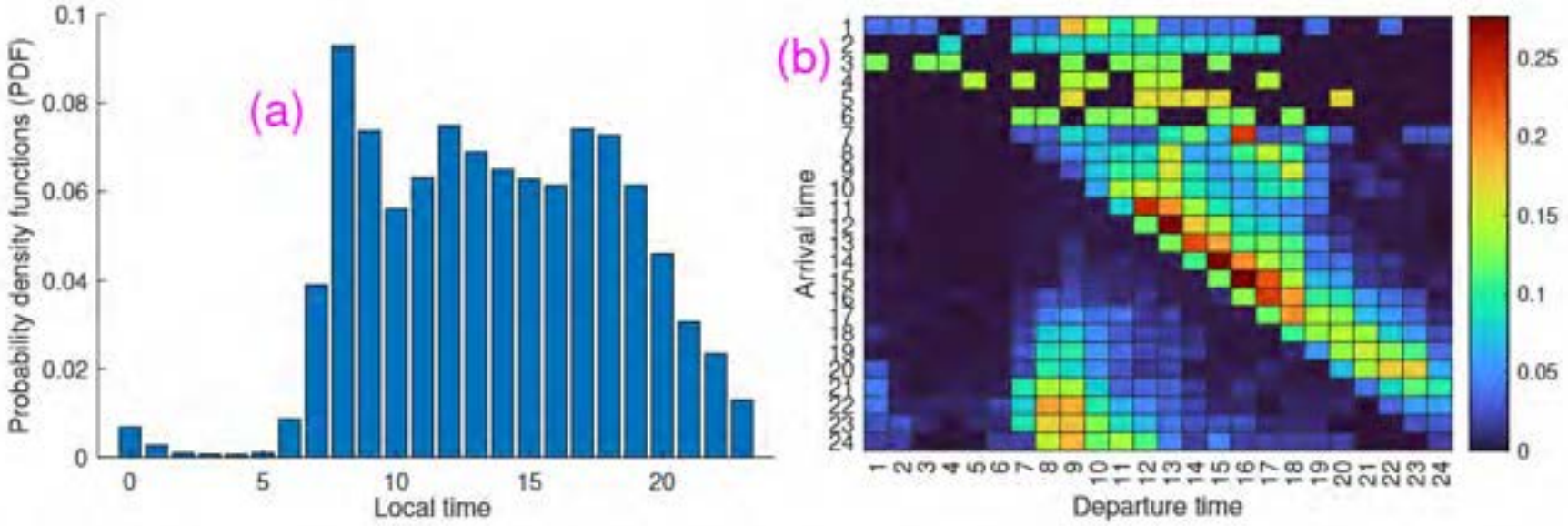
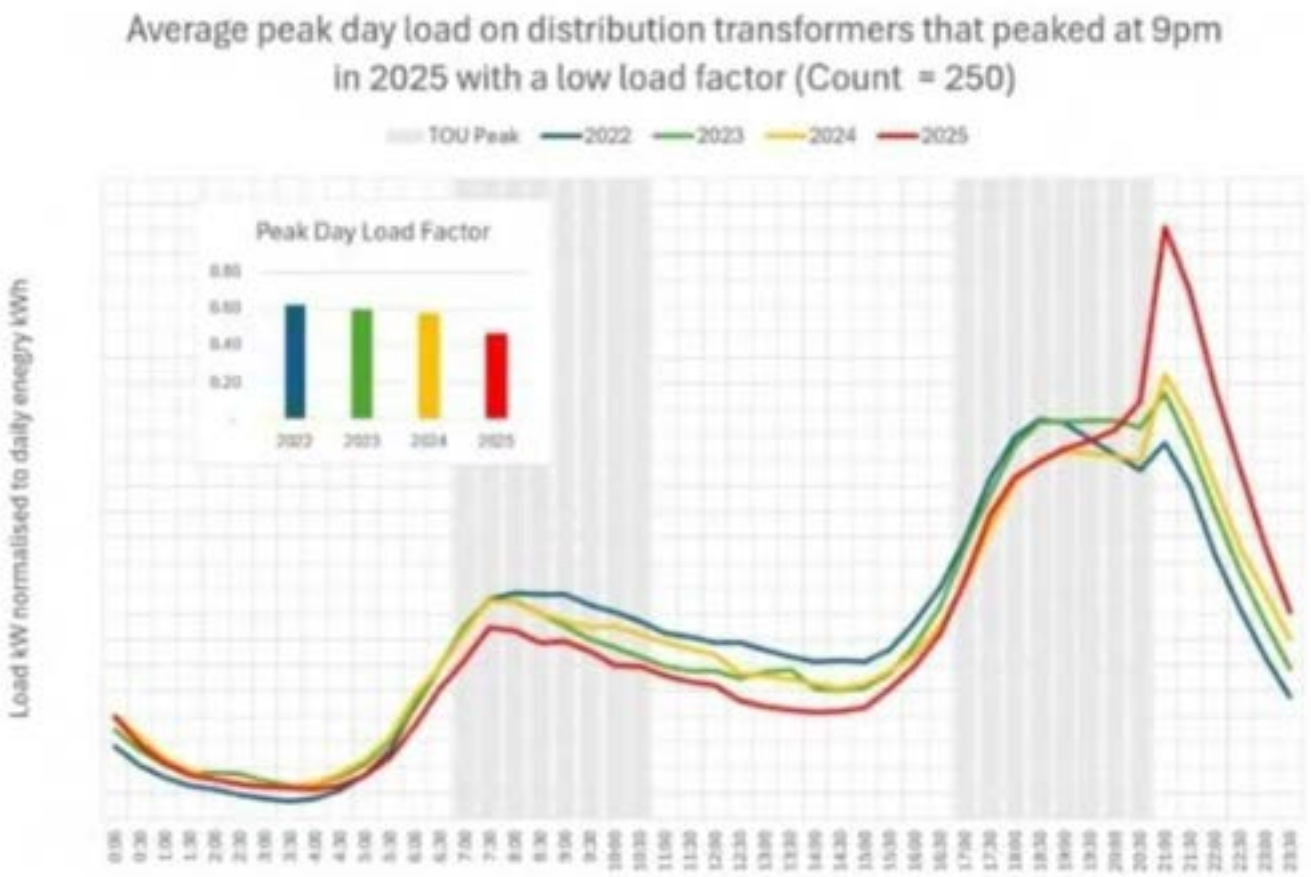


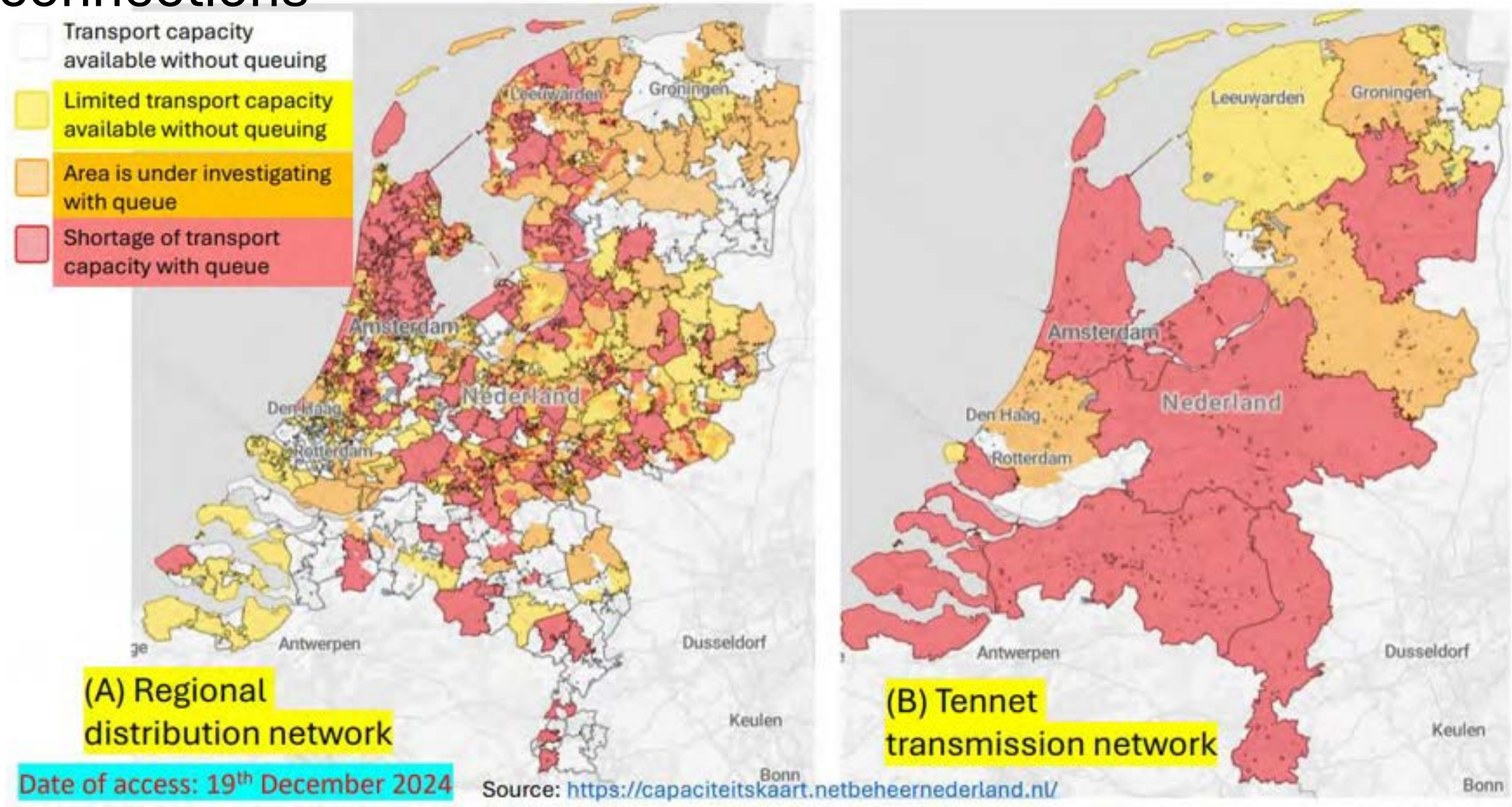
Fig. 1: Probability distribution t_{arr} (a) and conditional probability distribution $t_{dep} \mid t_{arr}$ (b).

3. Increased consumer rationality



Source: https://www.thepost.co.nz/business/360825632/kiwis-are-shifting-their-electricity-use-not-always-ways-are-helpful

4. Bottleneck in power networks – delaying new connections

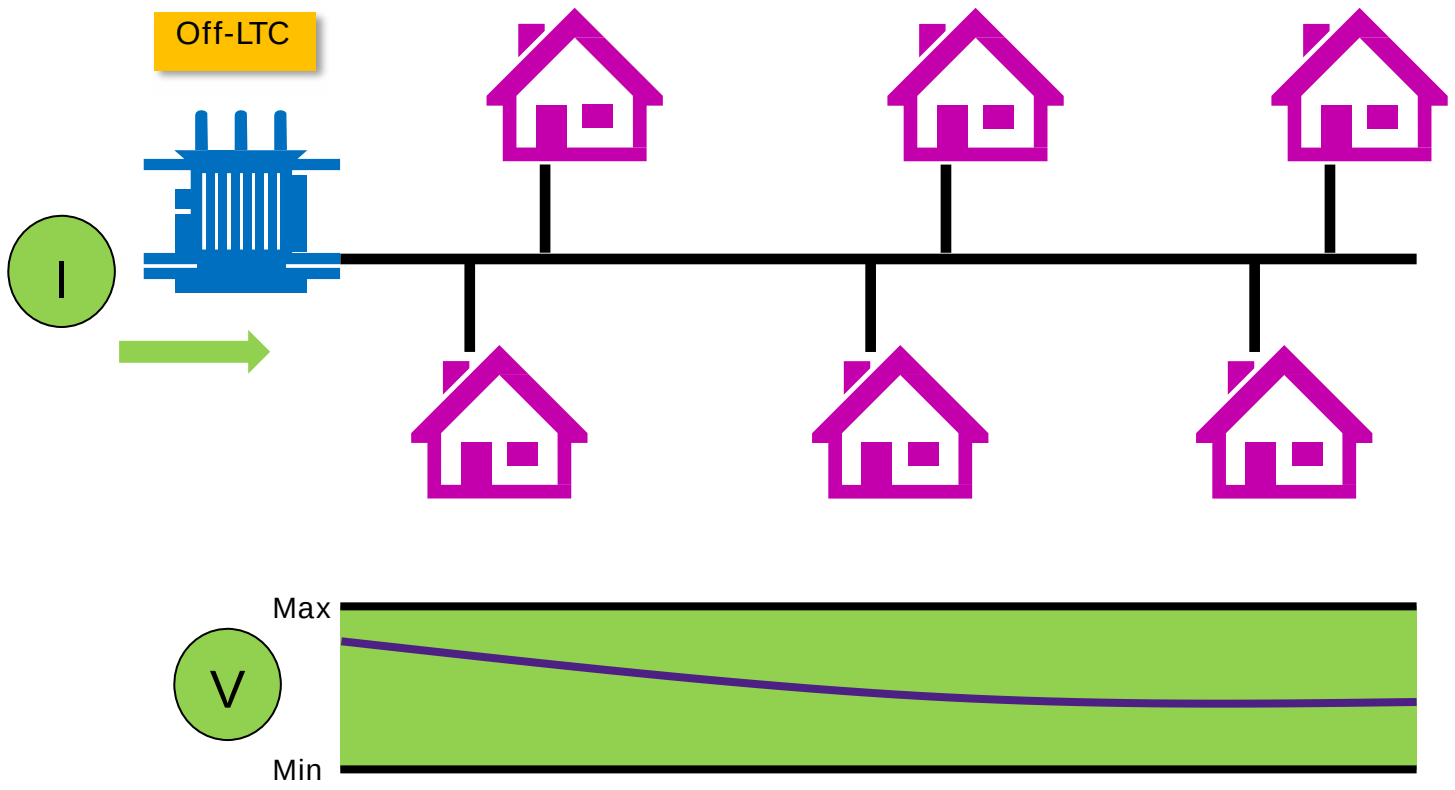


Date of access: 19th December 2024

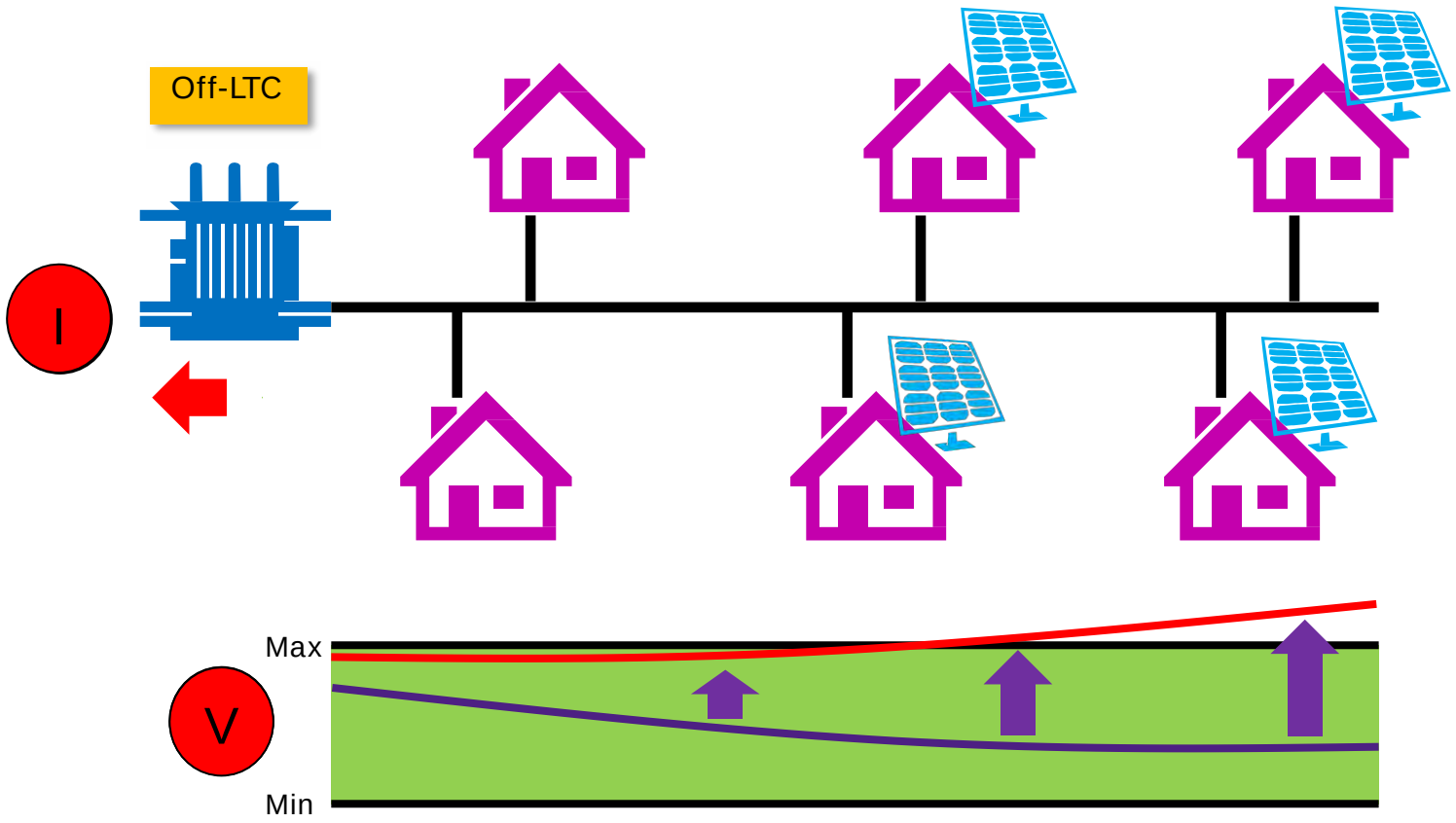
Source: https://capaciteitskaart.netbeheernederland.nl/

Illustrative example

Impact of DER integration

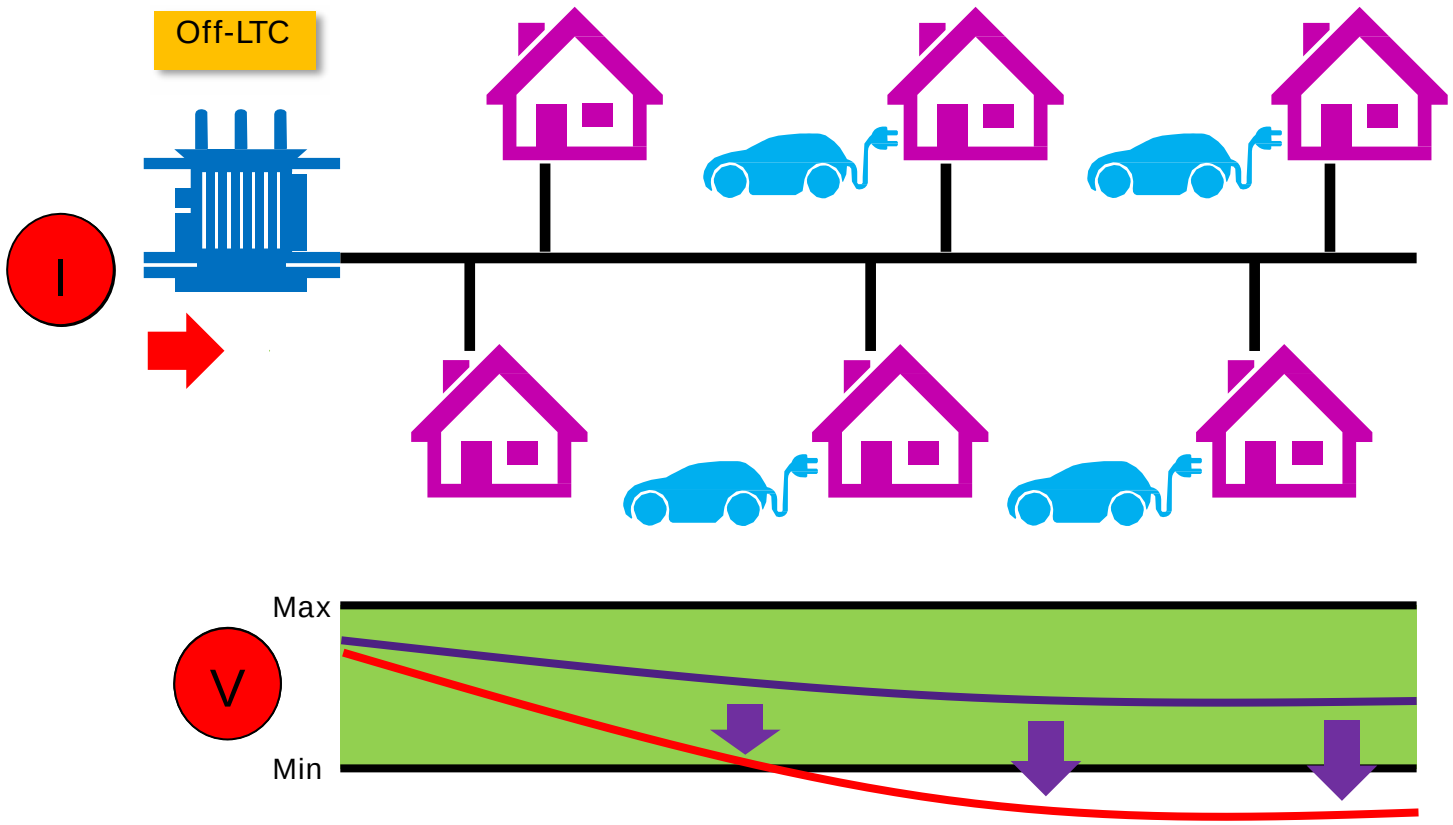


Graphic courtesy → Prof Luis(Nando) Ochoa (Uni. Of Melbourne) slides



PV generation happens during the day, when many people are not at home → Large reverse power flows → Voltage and thermal problems

Graphic courtesy → Prof Luis(Nando) Ochoa (Uni. Of Melbourne) slides



EVs are charged when people return home
→ Larger peak demand → Voltage and thermal problems

Graphic courtesy → Prof Luis(Nando) Ochoa (Uni. Of Melbourne) slides

What next?

The Operator's Decision

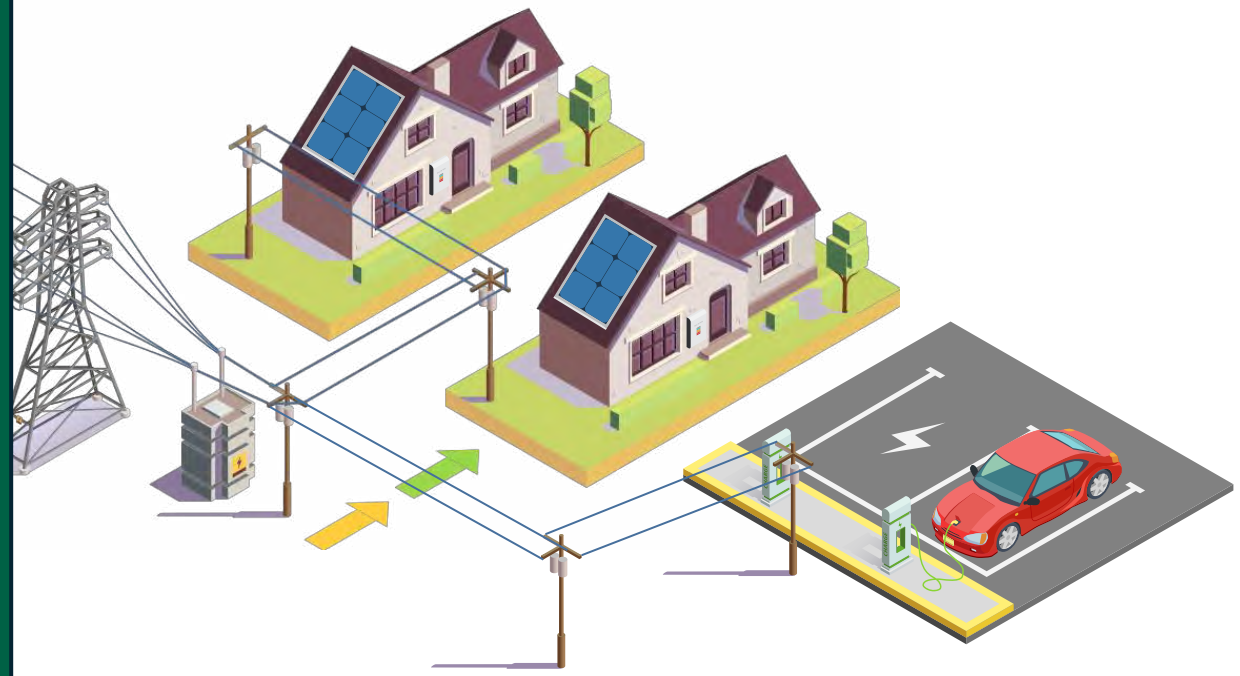
Hard vs Smart

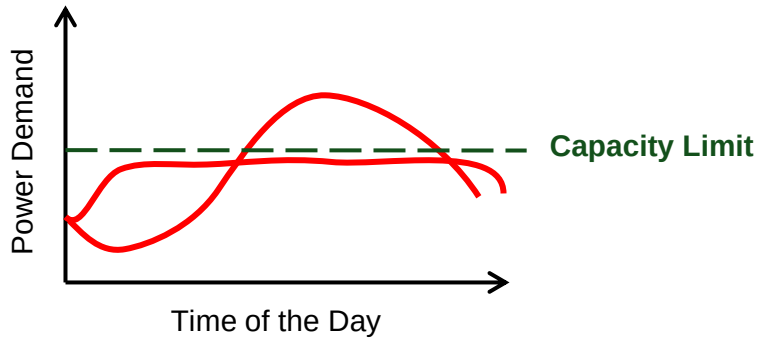
Network Reinforcement

Activating Flexibility

- ✗ Estimated **4 billions** of Euros for network reinforcement (just in Flanders)
- ✗ Project delays probably exceed climate goals
- ✓ **Save EUR 200 million** in Reinforcement Investments
- ✓ Meet climate goals on time

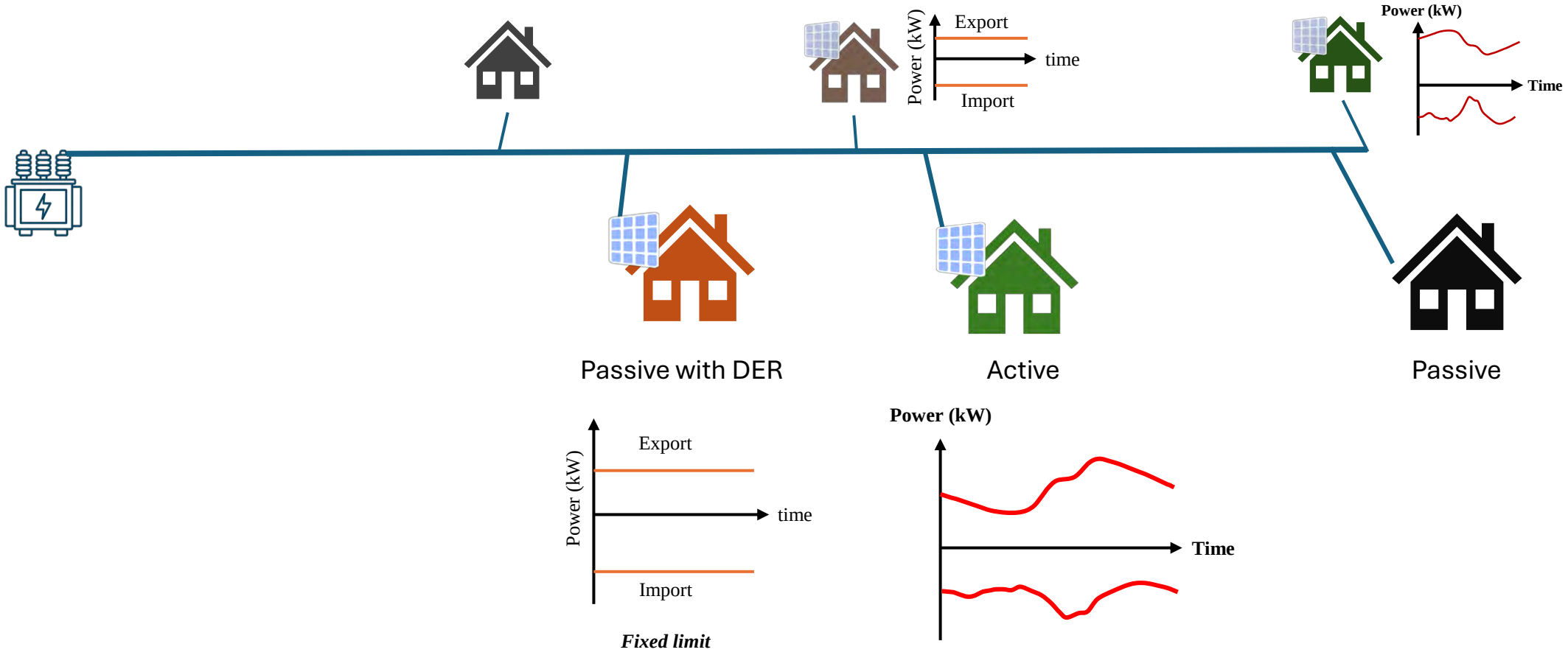
Fluvius 2024 annual report





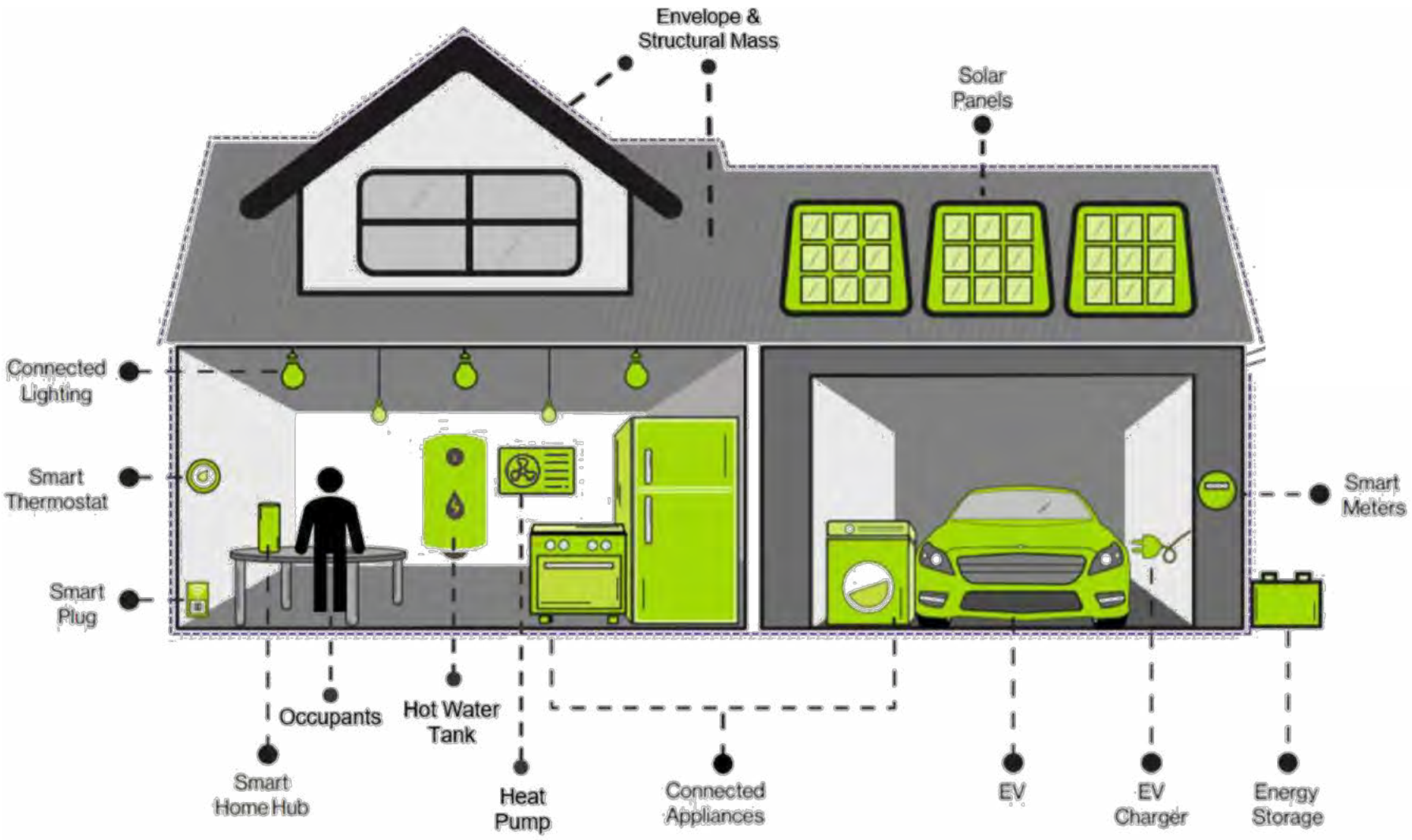
Slide courtesy: Mohammed Numair, KU Leuven

Active management and customer types



Slide courtesy: Ali Jalilian, KU Leuven

Active customer



Summary

- Power networks are evolving with increased volatility

- Aging power networks with higher uncertainty and simultaneity necessitates rethink secure operation of power networks
  - Distribution network incidents (DNIs) denote the incidents of power quality violations
    - Example voltage magnitude outside permissible range or line loading exceeding capacity

- DNIs are becoming more rampant and frequent compromising the operation and reliability.
  - DSOs need to plan for such instances to provide compensation or implement necessary network upgrades for avoiding DNIs.

Thank you for your attention!

Dr. Md Umar Hashmi
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Heverlee, 08.02.2023

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Summary

- With transition in power network Power quality metrics are deteriorating
  - Power quality is going to be growing in importance for all stakeholders!

- Distribution network incidents (DNIs) denote the incidents of power quality violations
  - Example voltage magnitude outside permissible range or line loading exceeding capacity

- DNIs are becoming more rampant and frequent compromising the operation and reliability.
  - DSOs need to plan for such instances to provide compensation or implement necessary network upgrades for avoiding DNIs.